# Beyond Tonnelier's conjectured bound: six cycles and uniform finiteness in continuous piecewise-affine Liénard systems

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**Abstract**

We study the continuous piecewise-affine Liénard system $\dot{x} = F(x) - y,\ \dot{y} = x$, and show that breakpoint complexity ensures uniform finiteness but does not give Tonnelier's conjectured upper bound. For an explicit continuous piecewise-affine function $F$ with rational parameters and three genuine breakpoints, we certify at least six distinct hyperbolic limit cycles of the corresponding system. All four affine parts are of focus type, and the six cycles are transverse to every switching line they meet and persist under small parameter perturbations. Five follow the same full switching itinerary, exhibiting multiple fixed points within one analytic return branch. Independently, for each fixed $n$, we prove a uniform finite bound on geometrically isolated periodic orbits for the entire family with at most $n$ breakpoints, including grazing cycles, with no restriction on the spectral type of the affine parts. The proof encodes each periodic orbit by at most $2n + 2$ affine flights and uniformly bounds each oscillatory phase, so that the resulting family of periodic-orbit markers is definable in $\mathbb{R}_{\mathrm{an,exp}}$. The resulting bound is non-effective. Finally, adapting an outer-breakpoint construction to the certified seed yields at least $n + 3$ hyperbolic limit cycles for every $n \geq 3$, with all slopes in $(-2, 2)$.

## 1 Introduction and main results

We consider

$$\dot{x} = F(x) - y, \qquad \dot{y} = x, \tag{1}$$

where $F : \mathbb{R} \to \mathbb{R}$ is continuous and piecewise affine. A genuine breakpoint is a point at which the adjacent slopes differ; it need not be a local extremum. A limit cycle is a geometrically isolated periodic orbit, counted without multiplicity. After translating $y$, we impose $F(0) = 0$, so the origin is the unique equilibrium. Let $H_c(n)$ be the supremum of the number of limit cycles among functions with exactly $n$ genuine breakpoints.

Two questions should be distinguished. The sharp counting problem asks how many limit cycles can occur at a prescribed number of breakpoints. The uniform finiteness problem asks whether their number is bounded independently of the coefficients and breakpoint positions once the number of breakpoints is fixed.

For the continuous class, Tonnelier proposed an upper bound by the number of breakpoints [1, Conjecture 2]. For two breakpoints, $H_c(2) = 2$ [2], and the construction of Chen–Li–Zhang–Zhang gives $H_c(n) \geq n$ for arbitrary $n$ [3]. For $n \geq 3$, the continuous upper-bound problem is stated as open in [3]. The present results separate the two roles of breakpoint complexity: the proposed estimate fails already at three breakpoints, whereas a bound depending only on that number exists over the entire parameter space.

Our first main result is a robust counterexample within the all-focus class. Our second establishes uniform finiteness over the full parameter space, independently of that construction. The third propagates the certified seed to larger breakpoint numbers.

**Theorem 1.1** (A robust six-cycle example)**.** *There is a continuous piecewise-affine function with rational parameters and exactly three genuine breakpoints whose system* (1) *has at least six distinct hyperbolic limit cycles. All four affine zones are of focus type, and the six cycles are transverse to every switching line they meet. Five of the six cycles follow the same full switching itinerary. These cycles, their switching itineraries and their stability persist on an open neighbourhood of the parameter point in the seven-dimensional normalised parameter space.*

**Theorem 1.2** (Uniform finiteness at fixed complexity)**.** *For every fixed integer $n \geq 0$, there is a finite integer $N_n$ such that every system* (1) *with a continuous piecewise-affine function $F$ having at most $n$ breakpoints has at most $N_n$ geometrically isolated periodic orbits. The bound is uniform in all slopes, intercepts, breakpoint positions and orbit amplitudes. It is non-effective in the present proof.*

**Theorem 1.3** (Lower bounds with more breakpoints)**.** *For every $n \geq 3$, there is a system with exactly $n$ genuine breakpoints, all slopes in $(-2, 2)$, and at least $n + 3$ distinct hyperbolic cycles transverse to all switching lines they meet. Hence*

$$n + 3 \leq H_c(n) \leq N_n < \infty \qquad (n \geq 3).$$

The results distinguish two sources of cycle complexity. A switching itinerary records the order of events, but enumerating itineraries does not bound the number of fixed points of an individual return branch. The five full-itinerary cycles make this distinction explicit within a continuous system with a unique equilibrium. At the same time, excursion geometry bounds the number of affine flights needed to describe a periodic orbit, and the location of the equilibria of the affine extensions bounds the oscillatory phase of each flight. These bounds yield uniform finiteness without a uniform bound on orbit amplitude or flight time.

Panazzolo's bounded-spiraling framework provides definability for flows restricted to suitable composable cell lists [5, Theorem 1.2]; his Section 5.5 formulates uniform finiteness questions for semi-limit cycles in continuous piecewise-linear families. The additional task here is to obtain a finite collection of descriptions that covers every periodic orbit uniformly over the full Liénard parameter space. We accomplish this by bounding both the number of flights and the oscillatory phase of each flight, including trajectories with grazing contacts. This yields a uniformly definable family of periodic-orbit markers and hence a uniform bound for geometrically isolated periodic orbits in this Liénard subclass.

Carmona, Fernández-Sánchez, and Novaes obtained the bound $L^* \leq 8$ for isolated crossing periodic solutions in two-zone planar piecewise-linear systems separated by a straight line [6]. Bravo and Trinidad-Forte proved finiteness for non-autonomous scalar piecewise-linear equations on a cylinder, with bounds depending on the number of regions and the degree of the trigonometric coefficients [7]. Here we treat continuous piecewise-affine Liénard systems with an arbitrary fixed number of breakpoints, including grazing periodic orbits.

Two further consequences concern the return gain and parameter dependence. Corollary 3.3 uses the alternating stability of the certified cycles to obtain extrema of the logarithmic return gain with alternating positive and negative values on the common branch. Corollary 4.5 gives a finite partition of parameter space with continuous ordered isolated-cycle markers and constant cycle count on each cell; the count changes only finitely often along each definable one-parameter path.

Section 2 fixes the section coordinates and affine-flow conventions. Section 3 gives the exact system, the shooting certificate, persistence and the common analytic branch. Section 4

encodes every periodic orbit by a bounded list of affine flights and applies the uniform fibre theorem for $\mathbb{R}_{\mathrm{an,exp}}$. Section 5 adapts the outer-breakpoint construction of Chen–Li–Zhang–Zhang [3], preserving hyperbolicity and transversality in the regime $|m_j| < 2$. Appendix A derives the action identity on the full-itinerary branch; Appendix B lists the interval input.

## 2 Excursion geometry and affine flight maps

We first record the geometric properties used in both main proofs. The upper and lower axis intersections provide unique orbit markers, while the bound on contacts with vertical lines controls the length of the later flight coding.

For a periodic orbit write its upper and lower intersections with the $y$-axis as $(0, s)$ and $(0, -r)$, respectively. Let $P_+(s)$ and $Q(r)$ denote the first return maps on the upper and lower half-axes, respectively. The following elementary fact is the geometric basis for both the shooting labels and the finite coding.

**Lemma 2.1** (Axis sections and vertical contacts)**.** *Every nonconstant periodic orbit has exactly one intersection with each open half of the $y$-axis, one positive maximum and one negative minimum of $x$, and at most two contacts with any vertical line, counting a tangency once.*

*Proof.* The orbit is a Jordan curve by uniqueness. Since $\int_0^T x(t)\,dt = 0$, it visits both half-planes. Away from the origin, the axis is crossed transversely because $\dot{x} = -y$. On an oriented Jordan curve, transverse intersections, ordered along the $y$-axis, alternate between entering and leaving the interior. All lower-axis crossings have the same direction, and so do all upper-axis crossings; hence each open half-axis contains at most one intersection. Since the orbit visits both half-planes, crossings in both directions occur, and hence each open half-axis is met exactly once.

At an $x$-critical point $x(t_0) = b \neq 0$, one has $y(t_0) = F(b)$. Since $\dot{x}(t_0) = 0$, we have $x(t_0 + \tau) - b = o(\tau)$. Local Lipschitz continuity of $F$, together with $\dot{y}(t_0) = b$, then gives, also when $b$ is a breakpoint,

$$\dot{x}(t_0 + \tau) = -b\tau + o(\tau), \qquad x(t_0 + \tau) = b - \tfrac{1}{2}b\tau^2 + o(\tau^2).$$

Thus $\dot{x}$ crosses strictly from positive to negative at a contact on the positive excursion and in the reverse direction on the negative excursion. There can be only one of each; otherwise a second crossing would require the opposite direction in between. The same sign argument gives the vertical-line bound, including at a breakpoint. □

For ordered breakpoints $\xi_1 < \cdots < \xi_n$ and slopes $m_0, \ldots, m_n$, use the continuous hinge parametrisation

$$F_\vartheta(x) = m_0 x + \sum_{j=1}^{n} (m_j - m_{j-1})\big((x - \xi_j)_+ - (-\xi_j)_+\big), \qquad u_+ = \max(u, 0). \tag{2}$$

The normalisation $F_\vartheta(0) = 0$ removes the additive constant in $F$, while the breakpoint positions and slopes remain free parameters. In a zone where $F(x) = mx + b$, the physical flow is

$$\Phi_{m,b}(t, z) = q + e^{A_m t}(z - q), \quad q = (0, b), \quad A_m = \begin{pmatrix} m & -1 \\ 1 & 0 \end{pmatrix}. \tag{3}$$

For velocity $v = F(x) - y$, $\dot{x} = v$, $\dot{v} = mv - x$ almost everywhere. In the oscillatory regime $|m| < 2$, with $a = m/2$, $\omega = \sqrt{1 - a^2}$,

$$\Psi_m^t = e^{at} \begin{pmatrix} \cos(\omega t) - a\sin(\omega t)/\omega & \sin(\omega t)/\omega \\ -\sin(\omega t)/\omega & \cos(\omega t) + a\sin(\omega t)/\omega \end{pmatrix}. \tag{4}$$

The velocity vector field can jump at a switch, while the physical vector field is continuous; variational calculations across switches are made in physical coordinates.

# 3 A robust counterexample with three breakpoints

We construct the six-cycle example by validating six zeros of suitable shooting equations. The same computation verifies their itineraries and hyperbolicity, after which persistence follows from the implicit-function theorem.

## 3.1 The exact system and the six certified cycles

In system (1), take $F = F_{\vartheta_*}$ as defined by (2), with

$$\begin{aligned}(\xi_1, \xi_2, \xi_3) &= (-4.568454409748787, -1, 4.697723054757359),\\ (m_0, m_1, m_2, m_3) &= (1.7852068867006083, -0.5628649147225554,\\ &\qquad 0.31155696868477084, -1.785205933274595).\end{aligned} \tag{5}$$

Every displayed decimal in (5) is an exact rational number. Denote this parameter point by $\vartheta_*$. The adjacent slopes are distinct and lie in $(-2, 2)$, so there are three genuine breakpoints and four focus zones.

Table 1 records the six certified cycles, labelled $\Gamma_1, \ldots, \Gamma_6$ in increasing upper-section coordinate $s_i^*$. The following subsections establish their existence, distinctness and stability. One cycle uses the short itinerary; five cycles traverse all four zones with the same full itinerary, specified in (7). Here $\chi$ denotes the divergence integral over one minimal period.

| cycle | upper coordinate (approx.) | strict interval for $\chi$ | stability |
|---|---|---|---|
| 1 | 5.7038049296 | $(-0.414496, -0.414495)$ | attracting |
| 2 | 5.9550028751 | $(0.040280, 0.040281)$ | repelling |
| 3 | 6.1572916585 | $(-0.009958, -0.009957)$ | attracting |
| 4 | 6.5817617853 | $(0.007095, 0.007096)$ | repelling |
| 5 | 18.7923449781 | $(-0.000234, -0.000233)$ | attracting |
| 6 | 168.8220183797 | $(0.00001595, 0.00001596)$ | repelling |

Table 1: Approximate section coordinates and rigorous enclosures of the characteristic exponents for the six cycles. Cycle 1 has the short itinerary; cycles 2–6 have the full itinerary.

Figure 1(a) shows the short orbit $\Gamma_1$ and the full-itinerary orbit $\Gamma_4$ in physical coordinates. All six cycles are shown in Figure 2, Appendix C.

## 3.2 Shooting equations and interval validation

For a list of prescribed nodes $a_0, \ldots, a_k$ and slopes $\sigma_1, \ldots, \sigma_k$, start with $v_0 = -s$ and set

$$(X_j, v_j) = \Psi_{\sigma_j}^{t_j}(a_{j-1}, v_{j-1}).$$

The unknown vector is $z = (s, t_1, \ldots, t_k)$, and the residual is

$$\mathcal{R}(z) = (X_1 - a_1, \ldots, X_k - a_k, v_k + s). \tag{6}$$

The short and full charts are

$$\begin{aligned}k = 3: &\quad (\sigma_j) = (m_2, m_1, m_2), \quad (a_j) = (0, \xi_2, \xi_2, 0),\\ k = 7: &\quad (\sigma_j) = (m_2, m_1, m_0, m_1, m_2, m_3, m_2),\\ &\quad (a_j) = (0, \xi_2, \xi_1, \xi_1, \xi_2, \xi_3, \xi_3, 0).\end{aligned} \tag{7}$$

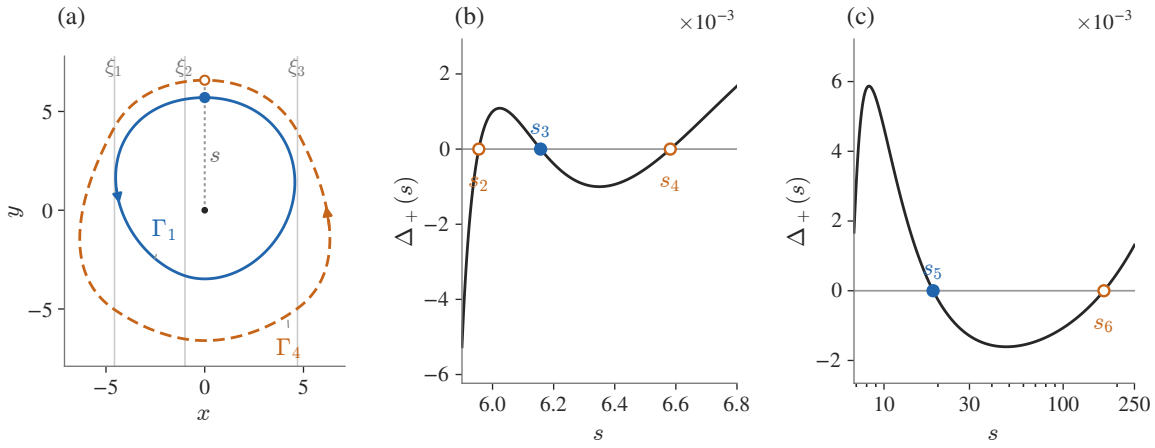



Figure 1: Numerical trajectories and upper-section return displacement for (5). (a) The short orbit $\Gamma_1$ and full orbit $\Gamma_4$ in physical coordinates. (b,c) The five certified full-itinerary zeros of $\Delta_+ = P_+ - s$; panel (c) uses a logarithmic horizontal axis. Blue solid curves and filled markers denote attracting cycles; orange dashed curves and open markers denote repelling cycles.

These shooting flights are split only at the prescribed nodes. A flight may cross the $y$-axis in its interior; the half-plane coding in Section 4 subdivides it there.

**Validation of the shooting zeros.** We use the following contraction criterion for interval validation [10, Section 13]. Let $X \subset B$ be nonempty closed bounded boxes in the same finite-dimensional space, let $z_0 \in B$, and suppose that $\mathcal{R}$ is $C^1$ on a neighbourhood of $B$. Let $\mathcal{J}$ enclose $D\mathcal{R}(z)$ for every $z \in B$, and let $C$ be a fixed square matrix. If

$$K \subset \operatorname{int} X, \qquad \sup_{J \in \mathcal{J}} \|I - CJ\|_\infty < 1, \tag{8}$$

where $K$ encloses

$$z_0 - C\mathcal{R}(z_0) + (I - C\mathcal{J})(B - z_0),$$

then $\mathcal{R}$ has a unique zero in $X$, lying in its interior. Indeed, for $T(z) = z - C\mathcal{R}(z)$ and $z \in X$, the segment from $z_0$ to $z$ is contained in $B$, so the integral mean-value formula gives $T(z) \in K$. The second inequality makes $T$ a contraction on $X$ and makes $C$ invertible by the Neumann series; fixed points of $T$ are therefore exactly zeros of $\mathcal{R}$.

For cycle $i$, the entry $i$ of `krawczyk_centers` in S1's `six-cycle-001.json` is an exact rational vector $c_i = (\bar{s}_i, \bar{t}_{i1}, \ldots, \bar{t}_{ik_i})$, with $k_1 = 3$ and $k_i = 7$ for $i \geq 2$. The shooting box is $X_i = c_i + [-10^{-25}, 10^{-25}]^{k_i+1}$. It encloses the section coordinate and flight times, with all system parameters fixed at $\vartheta_*$. Arb real-ball arithmetic [11] supplies directed enclosures at 480-bit precision. For each exact box $X$, an outward enclosure $B$ contains its rational endpoints, $z_0$ is the exact dyadic midpoint of $B$, and the preconditioner $C$ is a fixed dyadic matrix. Automatic differentiation supplies $\mathcal{J}$, and strict inclusion is checked against the exact rational endpoints of $X$.

**Validation of the prescribed trajectories.** For every flight, interval enclosures on 512 closed time cells cover the entire normalised interval $[0, 1]$. On a monotone flight, a strict velocity sign verifies monotonicity. On a flight returning to the same switch, initial and final blocks have outward and inward velocity, respectively; the remaining cells lie strictly on the

required side. Endpoint blocks use the exact endpoint equations, forward at departure and backward at arrival. The final flight of the short chart starts at $\xi_2$, crosses the lower half-axis, turns in zone 2, and reaches the upper-axis section with $v < 0$. Its initial block has $v > 0$; all remaining cells have $x > \xi_2$, and every cell has $x < \xi_3$. The other short turning flight also stays to the right of $\xi_1$, so the short orbit reaches neither outer breakpoint. Every switching velocity is separated from zero. The minima $d_{\min}$ and $u_{\min}$ below range only over the cells where the corresponding position or velocity inequality is imposed. The certified bounds are

| quantity | notation | strict bound |
|---|---|---|
| contraction defect | $q_{\max}$ | $< 7.3 \cdot 10^{-17}$ |
| inclusion margin | $\eta_{\min}$ | $> 9.99 \cdot 10^{-26}$ |
| position clearance | $d_{\min}$ | $> 0.0489$ |
| signed velocity | $u_{\min}$ | $> 0.000614$ |
| switching speed | $v_{\min}$ | $> 0.8084293584$ |

Table 2: Bounds used by the six-cycle interval certificate.

Let

$$z_i^* = (s_i^*, t_{i1}^*, \dots, t_{ik_i}^*) \in \operatorname{int} X_i$$

be the unique zero certified by (8). Its upper-section coordinate is recorded in Table 1. The upper-axis boxes are disjoint, so Lemma 2.1 gives six distinct geometric orbits.

**Proposition 3.1** (Certified periodic-orbit data)**.** *For the exact seed* (5)*, each of the six boxes $X_i$ defined above contains a unique zero of its shooting residual. The corresponding flight segments satisfy the prescribed first-event and switching-velocity inequalities, and the six upper-section coordinates are pairwise distinct. The first box uses the short chart and the other five use the same full chart.*

The interval construction above proves the proposition: strict inclusion and contraction give the six zeros, while the cell checks and endpoint blocks identify each zero with a complete trajectory.

### 3.3 Hyperbolicity and persistence

Each certified closed trajectory crosses $x = \xi_2$ exactly twice, and the first-event inequalities exclude any further contact with this line. Its primitive periodic orbit crosses the same line transversely, so it has at least two crossings; Lemma 2.1 gives at most two. The shooting trajectory therefore traverses one minimal period. In particular, the sum of its prescribed flight times is the minimal period, and the certified divergence integral is its characteristic exponent.

At a transverse switch the physical vector field is continuous, so the saltation matrix is the identity. The variational fundamental matrices therefore compose without an additional jump factor. In physical coordinates, Liouville's formula and the tangent multiplier one give

$$\mu = \exp(\chi). \tag{9}$$

The strict intervals in Table 1 avoid zero, so all six cycles are hyperbolic.

All event equations are analytic at the certified crossings. Their time derivatives are the nonzero crossing velocities, so the analytic implicit function theorem continues the flight times in the seven normalised parameters. If $P_i(s, \vartheta)$ denotes the continued local return, then at the seed

$$P_i(s_i^*, \vartheta_*) - s_i^* = 0, \qquad \partial_s(P_i - s)(s_i^*, \vartheta_*) = \mu_i - 1 \neq 0.$$

A second application of the theorem continues the six roots in disjoint section intervals. Intersecting the six parameter neighbourhoods and shrinking it if necessary keeps the slopes in $(-2, 2)$, the breakpoints ordered and genuine, and the multipliers on their original sides of one. This proves Theorem 1.1.

### 3.4 Coexisting cycles on one analytic return branch

The five full-itinerary cycles lie on a common analytic return branch. The essential point is that increasing the section amplitude increases both excursions, so the conditions for visiting all switches hold on one interval.

**Lemma 3.2** (The common full-itinerary branch)**.** *For the seed* (5)*, let $J$ be the set of upper-section amplitudes for which the first-return trajectory crosses all three switching lines twice and transversely. Then $J$ is an open interval containing $[s_2^*, s_6^*]$, and $P_+$ is analytic on $J$.*

*Proof.* Put $M = \max_j |m_j| < 2$. In $(x, v)$ coordinates the polar angle obeys, almost everywhere,

$$\dot\theta = -1 + \frac{m(x)xv}{x^2+v^2}, \qquad -1 - \frac{M}{2} \le \dot\theta \le -1 + \frac{M}{2} < 0.$$

Every axis half-return is therefore finite. Denote the right half-return from $(0, -r)$ to $(0, R(r))$ in physical coordinates by $R(r) > 0$. Since $\dot y = x > 0$ on this half-arc, it is a graph $x = h_r(y) > 0$ between its endpoints. Distinct half-arcs cannot meet by uniqueness, including at a common upper endpoint. If $r_2 > r_1$, then $h_{r_2}(-r_1) > h_{r_1}(-r_1) = 0$. Were $R(r_2) < R(r_1)$, the ordering would reverse at $y = R(r_2)$, forcing an intersection. Thus $R(r_2) > R(r_1)$, and $h_{r_2} > h_{r_1}$ throughout the smaller graph's domain. In particular its maximal excursion $M_R(r) = \max h_r$ is strictly increasing.

Applying the same argument to the reflected system with $\widetilde F(X) = -F(-X)$, obtained by $(X, Y) = (-x, -y)$, gives a strictly increasing left half-return $L(s)$ from $(0, s)$ to $(0, -L(s))$, with strictly increasing excursion $M_L(s)$. These functions are continuous by continuous dependence and the transverse axis crossings. The upper return is $P_+ = R \circ L$, so both $M_L(s)$ and $M_R(L(s))$ increase strictly. Each half-arc has exactly one turning point: at $v = 0$, $\dot v = -x$, so all such crossings have the same direction. Consequently a negative switch is crossed twice transversely precisely when $M_L(s) > |\xi_i|$, and a positive switch precisely when $M_R(L(s)) > \xi_i$. Each condition defines an open terminal interval; their intersection is the full-itinerary interval $J$ containing the five certified full orbits. On $J$ the six switch hits and the axis return are transverse. At each hit the time derivative of the event equation is its nonzero crossing velocity. The seven affine hitting maps, their crossing velocities and $P_+$ are therefore analytic by the implicit-function theorem. □

**Corollary 3.3** (Robust alternating gain extrema)**.** *There are an open neighbourhood $V$ of $\vartheta_*$ and an interval $K \subset (0, \infty)$ on which the full-itinerary upper return $P_+(s, \vartheta)$ is analytic for $\vartheta \in V$. The continued roots $s_2(\vartheta) < \cdots < s_6(\vartheta)$ lie in $K$. For every $\vartheta \in V$, put $G(s, \vartheta) = \log(P_+(s, \vartheta)/s)$. There exist points $u_j \in (s_{j+1}(\vartheta), s_{j+2}(\vartheta))$, $1 \le j \le 4$, such that*

$$\partial_s G(u_j, \vartheta) = 0, \qquad (-1)^{j+1} G(u_j, \vartheta) > 0. \tag{10}$$

*The critical points are chosen separately for each parameter value and need not be nondegenerate.*

*Proof.* By Lemma 3.2, the full-itinerary domain $J$ is an open interval containing $[s_2^*, s_6^*]$. Choose a compact interval $[a, b] \subset J$ with these roots in its interior. Its switch hits and axis returns are transverse. The analytic event equations and their first-event conditions therefore persist on a common parameter neighbourhood by compactness. After shrinking

this neighbourhood, they define the full-itinerary return on $K = (a, b)$, with all five continued roots inside $K$.

At each root, the return multiplier gives

$$\partial_s G(s_i(\vartheta), \vartheta) = \frac{\partial_s P_+(s_i(\vartheta), \vartheta) - 1}{s_i(\vartheta)} = \frac{e^{\chi_i(\vartheta)} - 1}{s_i(\vartheta)}, \qquad 2 \le i \le 6. \tag{11}$$

Table 1 and persistence give the signs $(+, -, +, -, +)$. For $j = 1, 3$, $G$ is positive immediately to the right of $s_{j+1}(\vartheta)$, while both endpoint values on $[s_{j+1}(\vartheta), s_{j+2}(\vartheta)]$ vanish. A positive maximum is therefore attained at an interior critical point. For $j = 2, 4$, the same argument gives an interior negative minimum. This proves (10). Additional zeros between the specified roots do not affect this argument. □

The common itinerary fixes event order but not residence times. If $t_1, \dots, t_7$ are the full-flight times, the divergence integral along one return is

$$\chi(s) = m_2(t_1 + t_5 + t_7) + m_1(t_2 + t_4) + m_0 t_3 + m_3 t_6. \tag{12}$$

Away from a fixed point this is a return divergence integral; at a fixed point it is the characteristic exponent. On $J$, the logarithmic return gain admits the decomposition

$$G(s) = \log \frac{P_+(s)}{s} = 2\pi q(m_2) + A_1(s) + A_2(s) + A_3(s), \qquad q(m) = \frac{m}{\sqrt{4 - m^2}}.$$

Here $A_i(s)$ is the net invariant jump at the two crossings of the $i$-th switching line. Appendix A defines these actions and derives the identity and their variation directions $(+, -, -)$ for (5). The action identity identifies competing amplitude-dependent contributions along the common itinerary. The certified roots establish that these contributions balance repeatedly; their individual variation signs alone do not force the observed number of roots.

Sharp counting therefore requires control of the zeros within individual itinerary branches. The next section obtains a uniform bound for isolated periodic orbits through a finite encoding of their trajectories.

# 4 Uniform finiteness from bounded-flight coding

Fix $n$. Use the parameter space

$$\mathcal{P}_n = \{(\xi_1, \dots, \xi_n, m_0, \dots, m_n) : \xi_1 < \cdots < \xi_n\}$$

and allow zero slope jumps; this represents systems with fewer than $n$ genuine breakpoints by inserting auxiliary slots.

The proof has two geometric inputs: a uniform bound on the number of affine flights in one periodic orbit, and a uniform bound on the oscillatory phase of each flight. We encode these flights with first-event conditions and prove that the codes describe exactly the periodic orbits. Their unique lower-axis markers then form a definable family, to which uniform finiteness applies.

In this section, a coding flight is a segment between consecutive contacts with the breakpoint lines or the $y$-axis. Unlike the shooting flights of Section 3, each such segment lies in a fixed open half-plane except at its endpoints. For the certified seed, inserting the intermediate lower-axis crossing refines the three- and seven-flight shooting charts into four- and eight-flight codes, respectively.

**Lemma 4.1** (Uniform phase bound for half-plane flights)**.** *Let $|m| < 2$, $\omega = \sqrt{1 - m^2/4}$, and $T > 0$. Suppose a nonconstant trajectory of the affine field $\dot{x} = mx + b - y$, $\dot{y} = x$ remains in one open half-plane $x > 0$ or $x < 0$ for $0 < t < T$, with both endpoints in its closure. Then $\omega T \le \pi$.*

*Proof.* The equilibrium of this affine extension is $(0, b)$. At phase $\pi$, the linear flow satisfies

$$e^{A_m \pi/\omega} = -e^{m\pi/(2\omega)} I.$$

If $x(0) \neq 0$, this gives $x(\pi/\omega) = -e^{m\pi/(2\omega)} x(0)$, which lies in the opposite half-plane. Thus $T < \pi/\omega$. If $x(0) = 0$, put $v_0 = b - y(0)$. Nonconstancy gives $v_0 \neq 0$, and

$$x(t) = e^{mt/2} \frac{v_0}{\omega} \sin(\omega t).$$

The first positive-time axis contact has phase $\pi$, so $T \le \pi/\omega$ in this case as well. □

This is a bound on phase, not on flight time: $\pi/\omega$ need not remain bounded as $|m| \to 2$. On the domains needed for these half-plane flights, the flow in a zone has the finite spectral form

$$e^{A_m t} = \begin{cases} e^{at}\{\cos(\omega t) I + t \operatorname{sinc}(\omega t) B_m\}, & m^2 < 4, \quad 0 \le \omega t \le \pi, \\ e^{at}(I + t B_m), & m^2 = 4, \\ e^{at}\{\cosh(\rho t) I + t \operatorname{sinhc}(\rho t) B_m\}, & m^2 > 4, \end{cases} \tag{13}$$

Here $t \ge 0$, $a = m/2$, $B_m = A_m - aI$, $\omega = \sqrt{1 - a^2}$ when $|m| < 2$, and $\rho = \sqrt{a^2 - 1}$ in the node case. The quotients $\operatorname{sinc} u = \sin u/u$ and $\operatorname{sinhc} u = \sinh u/u$ take the value one at zero. Lemma 4.1 restricts the trigonometric arguments to $[0, \pi]$. Restricted analytic functions and the exponential therefore make the resulting partial flow definable in $\mathbb{R}_{\mathrm{an,exp}}$, including the repeated-root and node cases. Together with the flight-count bound from Lemma 2.1, this is the geometric step needed for the following proposition.

**Proposition 4.2** (Definability of periodic-orbit markers)**.** *For fixed $n$, the set*

$$\mathcal{Z}_n = \{(\vartheta, r) \in \mathcal{P}_n \times (0, \infty) : (0, -r) \text{ lies on a nonconstant periodic orbit of } F_\vartheta\}$$

*is definable in $\mathbb{R}_{\mathrm{an,exp}}$, uniformly in all slopes, ordered breakpoint positions, and amplitudes.*

We prove the proposition by encoding periodic orbits with the following cell and first-event predicates. The construction uses the phase-restricted affine flows in (13). Let $I_0 = (-\infty, \xi_1)$, $I_j = (\xi_j, \xi_{j+1})$ for $1 \le j < n$, and $I_n = (\xi_n, \infty)$, with the evident convention for $n = 0$. Put

$$C_{j,\varepsilon}(\vartheta) = \{(x, y) : x \in I_j,\ \varepsilon x > 0\}, \qquad \varepsilon \in \{-1, +1\}.$$

For fixed parameters its closure is expressed inside the same definable structure by

$$z \in \overline{C_{j,\varepsilon}(\vartheta)} \iff \forall \delta > 0\ \exists w \in C_{j,\varepsilon}(\vartheta) \quad \|w - z\|^2 < \delta^2.$$

Write $\mathrm{Flow}_j(\vartheta; t, z, w)$ for the graph of the appropriate branch of (3) and (13). Endpoint matching alone is insufficient: an affine segment may leave its assigned zone before reaching the prescribed endpoint. We therefore require every intermediate point to remain in the corresponding open cell. Define $\mathrm{Flight}_{j,\varepsilon}(\vartheta; z, z', t)$ by

$$\begin{gathered} t > 0, \quad z, z' \in \overline{C_{j,\varepsilon}(\vartheta)}, \quad \mathrm{Flow}_j(\vartheta; t, z, z'), \\ \forall u\ \big(0 < u < t \Longrightarrow \exists w\, [\mathrm{Flow}_j(\vartheta; u, z, w)\ \wedge w \in C_{j,\varepsilon}(\vartheta)]\big). \end{gathered} \tag{14}$$

The final-time spectral condition makes all intermediate times admissible; the last line is the first-event guard and ensures that no switching line or the axis is met in the interior of a flight.

By Lemma 2.1, a periodic orbit records at most two contacts with each breakpoint line and one upper-axis crossing. Coincident events are merged, so the number $M$ of positive-duration flights satisfies

$$M \le 2n + 2. \tag{15}$$

Let $\mathcal{B}_\vartheta$ be the union of the axis and the breakpoint lines. Fix integers $2 \le M \le 2n+2$, $1 \le h < M$, and $0 \le j_i \le n$. A code consists of $r, s > 0$, times $t_i > 0$, and nodes $z_0, \dots, z_M$, satisfying

$$\begin{gathered} r > 0, \quad z_0 = z_M = (0, -r), \quad z_h = (0, s), \\ z_i \in \mathcal{B}_\vartheta \quad (1 \le i < M), \qquad x(z_i) = 0 \iff i = h \quad (1 \le i < M), \\ \mathrm{Flight}_{j_i, \varepsilon_i}(\vartheta; z_{i-1}, z_i, t_i) \quad (1 \le i \le M), \\ \varepsilon_i = +1 \ (i \le h), \qquad \varepsilon_i = -1 \ (i > h). \end{gathered} \tag{16}$$

Here $h$ marks the upper-axis crossing and $j_i$ identifies the zone of the $i$-th flight. There are finitely many choices of these indices. Projecting their finite union to $(\vartheta, r)$, eliminating the times, intermediate nodes and upper coordinate, gives a definable set. The next lemma identifies this projection with $\mathcal{Z}_n$.

**Lemma 4.3.** *The pair $(\vartheta, r)$ admits a code (16) if and only if $(0, -r)$ lies on a nonconstant periodic orbit.*

*Proof.* A periodic orbit supplies its contact list and hence a code by (15). Consecutive events have positive separation; the intervening trajectory lies in one open cell, and its oscillatory phase, when $|m| < 2$, is bounded by Lemma 4.1. The positive excursion precedes the negative one, as prescribed in (16).

Conversely, concatenate the flights of a code. The first-event guards identify each affine field with the global field on the open flight. At the endpoints, continuity of $F$ makes the derivatives agree with the physical field, including at cyclic closure. The concatenation is therefore a $C^1$ solution and uniqueness makes it the actual orbit. It has positive duration and avoids the equilibrium. There is no interior axis contact except the single upper-axis node, so it describes one circuit. Projection to the unique lower-axis coordinate identifies codes for the same orbit. □

**Boundary and degenerate cases.** A grazing contact separates two consecutive positive-duration flights in the same cell. A switch at zero is merged with the axis event and adds no extra node. Zero slope jumps merely insert auxiliary slots without changing the field. Empty cells contribute no flights because their fixed-parameter closures are empty, and $m^2 = 4$ uses the repeated-root branch of (13). Centres and period annuli contribute intervals of periodic markers, whose points are not isolated. The condition $r > 0$ and the open-cell guards exclude the equilibrium, also when it lies on a switch.

**Lemma 4.4** (Uniform fibre-component bound)**.** *For every definable family $Z \subset P \times \mathbb{R}$ in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$, there is an integer $B_Z$ such that every fibre $Z_p$ has at most $B_Z$ connected components [9, Section 4.4].*

*Proof.* Take a cylindrical cell decomposition compatible with $Z$, with the last coordinate as the fibre coordinate. The fibre of each cell is empty, a point, or an interval. Hence the total number of cells bounds the number of connected components of every fibre. □

*Proof of Proposition 4.2 and Theorem 1.2.* The finite definable union and projection above, together with Lemma 4.3, prove the marker proposition.

*From isolated markers to isolated periodic orbits.* For fixed parameters, geometric isolation of a periodic orbit is equivalent to isolation of its lower-axis marker in $(\mathcal{Z}_n)_\vartheta$. Indeed, $\dot{x} = r > 0$ at $(0, -r)$, so transversality and continuous dependence give a local return for nearby section coordinates, including when the given orbit grazes a switching line. Any nearby periodic coordinate must be a fixed point of this local return, since Lemma 2.1 excludes multiple lower-axis crossings in one circuit. Continuous dependence then makes the corresponding closed orbits close. Conversely, periodic orbits sufficiently close to the given orbit meet its transverse section at nearby coordinates. Thus accumulation of periodic markers and accumulation of periodic orbits are equivalent here. A continuous local return suffices for this argument.

The structure $\mathbb{R}_{\mathrm{an,exp}}$ is o-minimal [8, Corollary 5.13]; Lemma 4.4 gives a constant $B_n$ with

$$\#\pi_0((\mathcal{Z}_n)_\vartheta) \le B_n \quad \text{for every } \vartheta.$$

Consequently the number of isolated periodic orbits is at most $B_n$, which proves Theorem 1.2. For $n = 0$, the normalised linear system has no isolated cycles. □

### 4.1 Isolated markers and parameter dependence

**Corollary 4.5** (Finite parameter cells and definable scans)**.** *For fixed $n$, the parameter space $\mathcal{P}_n$ has a finite partition into definable cells $C$ such that each $C$ has an integer $k_C \ge 0$ and, when $k_C > 0$, continuous definable functions*

$$r_{C,1} < \cdots < r_{C,k_C} : C \longrightarrow (0, \infty)$$

*whose values are exactly the lower-axis markers of all geometrically isolated periodic orbits. In particular, their number is $k_C$ throughout $C$. Along any continuous $\mathbb{R}_{\mathrm{an,exp}}$-definable path $\gamma : T \to \mathcal{P}_n$, where $T \subset \mathbb{R}$ is an interval, this count is constant on a finite partition of $T$ into points and intervals. Neither the parameter space nor $T$ need be bounded.*

*Proof.* Define the fibre-isolated marker set $\mathcal{I}_n$ by

$$\begin{aligned}(\vartheta, r) \in \mathcal{I}_n \quad \Longleftrightarrow \quad & (\vartheta, r) \in \mathcal{Z}_n \ \text{ and } \ \exists \varepsilon > 0 \\ & \forall u > 0 \,\big[((\vartheta, u) \in \mathcal{Z}_n \ \wedge \ |u - r| < \varepsilon) \Longrightarrow u = r\big].\end{aligned} \tag{17}$$

It is definable by Proposition 4.2; the isolation equivalence just proved identifies its fibres with the isolated cycles. Each fibre is finite by Theorem 1.2. Apply cylindrical cell decomposition with the marker coordinate last [9, Section 4.2], compatibly with $\mathcal{I}_n$ and $\mathcal{P}_n \times \mathbb{R}$. Over a parameter cell $C$, no band cell can belong to $\mathcal{I}_n$, since it would have an interval fibre. The remaining cells are graphs of continuous definable functions on $C$. They are disjoint and hence can be ordered as displayed; they exhaust $\mathcal{I}_n$ over $C$. If there are none, $k_C = 0$. Finally, each $\gamma^{-1}(C)$ is definable in one real variable and thus is a finite union of points and intervals. Refining these finitely many sets proves the assertion about the path. □

In particular, the isolated-cycle count changes only finitely many times along any affine parameter path contained in $\mathcal{P}_n$. The cell decomposition is non-effective.

## 5 Propagating the six-cycle construction

We adapt the outer-breakpoint construction of Chen–Li–Zhang–Zhang [3, Section 3.1] to the certified six-cycle seed. The construction preserves a prescribed finite collection of hyperbolic cycles and adds an outer cycle; a common-slope perturbation, as in the rotated-family theory

[4], ensures hyperbolicity and transversality of the resulting family. Iterating the increment from the three-breakpoint seed yields the lower bound $H_c(n) \geq n+3$ for $n \geq 3$.

**Proposition 5.1** (Outer-breakpoint increment)**.** *Suppose that $F(0) = 0$, that $F$ has exactly $q$ genuine breakpoints, and that all its slopes lie in $(-2, 2)$. Given $k \geq 1$ hyperbolic cycles, there is another such function with exactly $q+1$ genuine breakpoints, nearby continuations of the chosen cycles, and an additional hyperbolic cycle outside them. All $k+1$ cycles can be made transverse to the switching lines they meet.*

## 5.1 Global return and gain at infinity

Use lower-axis coordinates and write $D_F(r) = Q_F(r) - r$. In velocity coordinates $(x, v)$, with $M = \max |m_j| < 2$,

$$\dot{\theta} = -1 + \frac{m(x)xv}{x^2+v^2} \leq -1 + M/2 < 0, \qquad \frac{d}{dt} \log \sqrt{x^2+v^2} = \frac{m(x)v^2}{x^2+v^2}. \tag{18}$$

The initial point $(x, y) = (0, -r)$ is $(x, v) = (0, r)$. Its radius cannot vanish in finite time, and its first full clockwise turn takes time between $2\pi/(1+M/2)$ and $2\pi/(1-M/2)$. Thus the lower return is defined for every $r > 0$.

Let $m_-, m_+$ be the tail slopes and define $F_\infty(X) = m_- X$ for $X \leq 0$ and $F_\infty(X) = m_+ X$ for $X \geq 0$. Since $F - F_\infty$ is bounded,

$$\sup_X |F(rX)/r - F_\infty(X)| \leq C_F/r \longrightarrow 0.$$

Under the physical scaling $(x, y) = r(X, Y)$, the vector fields have a common Lipschitz constant. Gronwall's inequality therefore gives convergence of the trajectories from $(0, -1)$ on bounded time intervals. Scaled velocities and lifted angles converge as well, and the strict angular bound in (18) makes the first full-turn times converge. The homogeneous half-plane of slope $m$ contributes the gain $\exp(\pi m/\sqrt{4-m^2})$, whence

$$\lim_{r\to\infty} \frac{Q_F(r)}{r} = \rho(m_-, m_+), \quad \rho(u, w) = \exp\!\left(\frac{\pi u}{\sqrt{4-u^2}} + \frac{\pi w}{\sqrt{4-w^2}}\right). \tag{19}$$

For fixed $u$, $\log \rho(u, w)$ increases continuously from $-\infty$ to $+\infty$ as $w$ traverses $(-2, 2)$.

## 5.2 Common-slope regularisation

**Lemma 5.2.** *Let $F_* : \mathbb{R} \to \mathbb{R}$ be continuous and piecewise affine, with $F_*(0) = 0$, finitely many fixed breakpoints, and slopes in $(-2, 2)$. For $F_\alpha = F_* + \alpha x$, restrict $\alpha$ to a neighbourhood of zero in which the slopes remain in $(-2, 2)$. The lower return $Q(r, \alpha)$ is defined for every $r > 0$, is jointly $C^1$, and satisfies* (20)*. Moreover:*

1. *If $D(r_-, 0)D(r_+, 0) < 0$, where $D(r, \alpha) = Q(r, \alpha) - r$ and $0 < r_- < r_+$, then for arbitrarily small $\alpha$ the interval $(r_-, r_+)$ contains a fixed point and every fixed point in $[r_-, r_+]$ is hyperbolic.*
2. *Any finite collection of hyperbolic cycles at $\alpha = 0$ has nearby hyperbolic continuations transverse to every switching line they meet for arbitrarily small suitable $\alpha$.*

*Proof.* Global return follows from (18).

**Step 1.** $C^1$ **regularity across grazing contacts.** On a finite non-equilibrium trajectory segment, a switch contact is transverse or has the quadratic expansion in Lemma 2.1; a contact on the axis is transverse. There are consequently only finitely many contacts on the segment. Write $z = (x, y)$ for a reference trajectory and $z_h$ for the trajectory with initial-state and parameter perturbation $h = (\delta z_0, \delta\alpha)$. Set $\delta z_h = z_h - z$, and define almost everywhere

$$\lambda(t) = F_*'(x(t)) + \alpha, \qquad L(t) = \begin{pmatrix} \lambda(t) & -1 \\ 1 & 0 \end{pmatrix}.$$

The remainder $R_h$ is specified by

$$\delta\dot{z}_h = L(t)\delta z_h + \delta\alpha(x(t), 0) + R_h(t), \qquad \delta z_h(0) = \delta z_0.$$

Continuous dependence gives $\delta z_h = O(\|h\|)$ uniformly on the segment. Let $E_h$ be the reference times at which $x(t)$ lies within $C\|h\|$ of a switch, with $C$ large enough to cover the state variation. Transverse and quadratic contacts give $|E_h| = O(\|h\|^{1/2})$. Outside $E_h$, both trajectories lie in the same affine zone, so only the product $\delta\alpha(x_h - x)$ contributes to $R_h$. The Lipschitz bound on $F_*$ yields

$$|R_h(t)| \le C\|h\|\mathbf{1}_{E_h}(t) + C\|h\|^2, \qquad \int |R_h(t)|\, dt = O(\|h\|^{3/2}) = o(\|h\|).$$

Variation of constants proves Fréchet differentiability. For nearby reference trajectories, the bounded variational coefficients converge almost everywhere off the finite contact set, hence in $L^1$. Their variational solutions therefore converge uniformly, proving continuity of the derivatives. The terminal lower-axis crossing is transverse, so the implicit-function theorem gives a jointly $C^1$ return.

**Step 2. Strict parameter monotonicity.** Let $X_\alpha(x, y) = (F_*(x) + \alpha x - y, x)$. For parameter variation $\eta = \partial_\alpha z$ with the initial point fixed, the wedge $W = \det(X_\alpha, \eta)$ satisfies $W' = \lambda W - x^2$ and $W(0) = 0$. At the return time, its value is $-QQ_\alpha$; variation of the return time adds a multiple of $X_\alpha$ and does not change the wedge. Applying the same argument to initial-section variation gives

$$\begin{aligned} Q_\alpha(r, \alpha) &= \frac{1}{Q(r, \alpha)} \int_0^T \exp\left( \int_t^T \lambda(u)\, du \right) x(t)^2\, dt > 0, \\ Q_r(r, \alpha) &= \frac{r}{Q(r, \alpha)} \exp\left( \int_0^T \lambda(t)\, dt \right) > 0. \end{aligned} \tag{20}$$

**Step 3. Hyperbolic fixed points for regular parameter values.** Since $D_\alpha = Q_\alpha > 0$, the zero set is locally a graph $\alpha = A(r)$. On the compact amplitude interval in the first assertion, the endpoint signs persist for a sufficiently small closed parameter interval. The zero set in this compact rectangle is covered by finitely many $C^1$ graphs $\alpha = A(r)$, and

$$A'(r) = -\frac{D_r(r, A(r))}{D_\alpha(r, A(r))}$$

on each graph. The critical values of a scalar $C^1$ function form a null set: on compact subintervals, cover the critical set by intervals where $|A'| < \varepsilon$, apply the mean value theorem, and then exhaust the domain. Choose $\alpha$ outside the finite union of these null sets. Opposite endpoint signs give a zero; every zero in the interval has $D_r = Q_r - 1 \neq 0$. Since $Q_r > 0$ and $Q_r \neq 1$, each such fixed point is hyperbolic.

**Step 4. Removal of grazing along hyperbolic branches.** For the second assertion, continue the chosen hyperbolic cycles on a common small parameter interval. Now $z(t,\alpha)$ follows a periodic orbit branch with period $T(\alpha)$, so its parameter variation does not have fixed initial value. Differentiating periodicity gives $\eta(t+T) = \eta(t) - T'X_\alpha(t)$. The wedge is therefore periodic and still satisfies $W' = \lambda W - x^2$. With $\chi = \int_0^T \lambda(t)\,dt$, it follows that

$$W(t) = \frac{\displaystyle\int_t^{t+T} \exp\left(\int_u^{t+T} \lambda(v)\,dv\right) x(u)^2\,du}{e^\chi - 1} \neq 0.$$

Here hyperbolicity ensures $e^\chi \neq 1$. At the unique positive or negative extremum $b(\alpha)$ of $x$, one has $X_\alpha = (0, b)$, hence $b' = -W/b \neq 0$. The derivative of this extremal value follows from uniqueness of the extremum and the uniform $C^1$ expansion on a fixed phase interval. On a sufficiently small continuation interval its sign is fixed. Each extremum can therefore equal any fixed switch position at most once. Avoiding finitely many such parameter values makes all chosen cycles transverse. □

### 5.3 Proof of the increment and the lower bound

*Proof of Proposition 5.1.* Order the lower-section coordinates of the selected cycles as $0 < r_1 < \cdots < r_k$. Hyperbolicity allows us to choose $r_0 > r_k$ sufficiently close to $r_k$ with $D_F(r_0) \neq 0$. Choose $R > 0$ larger than every existing breakpoint and every $x$-coordinate attained on the selected cycles or on the complete return trajectory from $r_0$. For $a \in (-2, 2)$, different from the old right-tail slope $m_+$, put

$$F_{R,a}(x) = F(x) + (a - m_+)(x - R)_+.$$

There is exactly one new genuine breakpoint. The selected cycles, their local return maps, and the complete return trajectory at $r_0$ are unchanged; in particular, $D_{R,a}(r_0) = D_F(r_0)$. Choose $a$ so that $\rho(m_-, a) - 1$ has sign opposite to $D_F(r_0)$. Equation (19) gives a finite $r_* > r_0$ such that

$$D_{R,a}(r_0) D_{R,a}(r_*) < 0.$$

Let $\sigma_j$ be the slopes of $F_{R,a}$ and put $\delta = \min_j(2 - |\sigma_j|) > 0$. For $|\alpha| < \delta/2$, the function $F_{R,a} + \alpha x$ keeps all slopes in $(-2, 2)$ and preserves every slope jump. Shrinking this parameter interval preserves both endpoint signs and continues the old cycles in disjoint section intervals below $r_0$. Lemma 5.2 first provides an additional hyperbolic root in $(r_0, r_*)$; a further small shift makes these $k+1$ cycles transverse while preserving their hyperbolicity and section order. By Lemma 2.1, disjoint periodic orbits are Jordan curves containing the origin, and their section order makes them nested. The additional cycle lies outside the continuations of the chosen cycles. □

Starting from Theorem 1.1, apply the proposition $n-3$ times. Every step retains genuine breakpoints and slopes in $(-2, 2)$, and increases the chosen number of hyperbolic transverse cycles by one, proving Theorem 1.3. At each step, the selected family contains three more cycles than the number of breakpoints, giving $6 + (n-3) = n + 3$.

## 6 Consequences and open questions

The breakpoint count does not give the conjectured upper bound for limit cycles, even within the continuous all-focus class. Five certified cycles coexist on one analytic return branch, showing that a fixed switching itinerary can support several distinct balances of contraction and expansion.

Nevertheless, fixed breakpoint complexity gives a uniform finite bound over the full parameter space. The decisive controls are the number of affine flights and their oscillatory phases, rather than uniform bounds on amplitudes or periods. The six-cycle example and the finiteness theorem therefore describe complementary aspects of the same counting problem. The marker theorem also gives finite variation of the isolated-cycle count along each definable parameter path.

The results leave open the sharp value of $H_c(3)$, effective bounds for $N_n$, and their growth with $n$. The present results give

$$6 \le H_c(3) \le N_3 < \infty, \qquad H_c(n) \ge n+3 \quad (n \ge 3).$$

Analytic conditions ensuring the coexistence of several cycles within one switching itinerary would complement the certified construction.

# A Action balance on the full-itinerary branch

Fix the all-focus seed (5), and let $J$ be the common full-itinerary interval from Lemma 3.2. The origin lies in zone 2, so the winding contribution uses $m_2$.

Write $p = v = F(x) - y$ for the velocity coordinate. In a zone of slope $m$, define

$$E_{m,x}(p) = p^2 - mxp + x^2, \qquad q(m) = \frac{m}{\sqrt{4-m^2}}.$$

Since $|m| < 2$, this energy is positive away from the origin. Remove the ray $\{(0,p) : p \le 0\}$ from the $(x,p)$-plane, and let $\phi_m(x,p)$ be the branch, with values in $(-5\pi/2, -\pi/2)$, of the argument of

$$\sqrt{4-m^2}\,x + i(2p - mx).$$

The preimage of the argument cut is the same removed ray for every $m$. Thus the values at both crossings of any switching line $x = \xi_i \neq 0$ use one branch, rather than independent flightwise lifts. Along a flight, $\dot{x} = p$ and $\dot{p} = mp - x$, whence

$$\dot{\phi}_m = -\frac{\sqrt{4-m^2}}{2}, \qquad \dot{E}_{m,x}(p) = mE_{m,x}(p).$$

Consequently

$$I_m(x,p) = \frac{1}{2}\log E_{m,x}(p) + q(m)\phi_m(x,p), \qquad \partial_p I_m(x,p) = \frac{p}{E_{m,x}(p)}, \tag{21}$$

and $I_m$ is constant on each flight. At the upper-axis launch $p = -s$, the trajectory enters $x < 0$ and the one-sided angle is $-\pi/2$; at its return $p = -P_+(s)$, it approaches from $x > 0$ with angle $-5\pi/2$. The intervening central flight crosses $x = 0$ with $p > 0$, away from the removed ray.

The switch $\xi_i$ separates slopes $m_{i-1}$ and $m_i$. Define

$$J_i(p) = I_{m_i}(\xi_i, p) - I_{m_{i-1}}(\xi_i, p).$$

Equation (21) gives

$$J_i'(p) = \frac{\xi_i(m_i - m_{i-1})p^2}{E_{m_{i-1},\xi_i}(p)E_{m_i,\xi_i}(p)}. \tag{22}$$

Let $p_i^+(s) > 0$ and $p_i^-(s) < 0$ be the rightward and leftward crossing velocities, respectively, and put

$$A_i(s) = J_i(p_i^+(s)) - J_i(p_i^-(s)).$$

Summing the invariant jumps over the six switch events leaves the endpoint change in zone 2. The map $(x,p)\mapsto(\sqrt{4-m_2^2}x, 2p-m_2x)$ has positive determinant and preserves the clockwise winding. The endpoint change is therefore $\log(P_+(s)/s)-2\pi q(m_2)$, so

$$G(s) := \log\frac{P_+(s)}{s} = 2\pi q(m_2) + A_1(s) + A_2(s) + A_3(s). \tag{23}$$

To determine the variation of each action, let $T_c(s)$ be the time of a transverse hit on a fixed switching line, and write $y_c(s)$ and $p_c(s)$ for its coordinates. The section-hit derivative is

$$\frac{dy_c}{ds} = \exp\left(\int_0^{T_c(s)} m(x(t;s))\,dt\right)\frac{-s}{p_c}, \qquad \frac{dp_c}{ds} = \exp\left(\int_0^{T_c(s)} m(x(t;s))\,dt\right)\frac{s}{p_c}.$$

Indeed, the flow determinant is the exponential shown, and projection onto the hit section divides by its nonzero normal velocity $p_c$. The second equality uses $p_c = F(\xi_i) - y_c$. Thus $p_i^+$ increases and $p_i^-$ decreases with $s$. Together with (22), this proves

$$\operatorname{sgn} A_i'(s) = \operatorname{sgn}[\xi_i(m_i - m_{i-1})].$$

For (5) these signs are $(+,-,-)$, giving competing amplitude-dependent contributions within the same itinerary.

For the derivative of the full return, let $a_i$ and $b_i$ be the slopes in the two zones adjacent to $\xi_i$, on the sides nearer to and farther from the origin, respectively. Thus $(a_i,b_i)=(m_i,m_{i-1})$ when $\xi_i<0$, and $(a_i,b_i)=(m_{i-1},m_i)$ when $\xi_i>0$. Let $p_i^{\mathrm{out}}$ and $p_i^{\mathrm{in}}$ denote the velocities pointing away from and toward the origin, and define

$$U_i = \frac{p_i^{\mathrm{out}}}{\xi_i} > 0, \qquad V_i = -\frac{p_i^{\mathrm{in}}}{\xi_i} > 0.$$

These normalisations apply on both sides: the outgoing velocity is $p_i^-$ on the negative side and $p_i^+$ on the positive side. Set

$$Q_i(s) = \frac{(U_i^2 - a_iU_i + 1)(V_i^2 + b_iV_i + 1)}{(U_i^2 - b_iU_i + 1)(V_i^2 + a_iV_i + 1)}, \tag{24}$$

which is positive. With $T(s)$ denoting the total return time, write $\chi(s)=\int_0^{T(s)} m(x(t;s))\,dt$, as in (12). Integrating $\dot E = mE$ on each of the seven flights and multiplying the resulting energy ratios cancels the intermediate endpoint energies and gives

$$e^{\chi(s)} = \left(\frac{P_+(s)}{s}\right)^2 \prod_{i=1}^{3} Q_i(s). \tag{25}$$

The section-hit derivatives likewise give $P_+'(s) = e^{\chi(s)}s/P_+(s)$. Hence, with $B(s) = \sum_i \log Q_i(s)$ and $\Delta_+(s) = P_+(s) - s$,

$$e^{B(s)} = 1 + sG'(s), \qquad \Delta_+'(s) = e^{G(s)+B(s)} - 1. \tag{26}$$

Thus $B=0$ locates critical points of $G$, rather than of $\Delta_+$ away from fixed points. At a fixed point $G=0$, the signs of $G'$, $\Delta_+'$, and $\chi$ agree.

# B Finite verification data

The interval input is supplied separately as Supplement S1, version 1.0.0. The archive name is

`S1-six-cycle-v1.0.0.zip`

Its top-level directory contains the following files:

- `six-cycle-001.json`;
- `certify_six_cycle_arb.py`;
- `certify_candidate_arb.py`;
- the directed report `six-cycle-480bit.json`.

The reference command is

```
python3 certify_six_cycle_arb.py six-cycle-001.json \
  --precision 480 --radius 1e-25 --slices 512 \
  --output certificate.json
```

Run this command inside the extracted top-level directory. The reference dependencies are `python-flint 0.9.0` and `mpmath 1.3.0`; S1's `ENVIRONMENT.md` and `requirements.txt` give the environment details. The command writes `certificate.json` and leaves the supplied reference report unchanged. All decimal inputs are interpreted as exact rational numbers. Approximate calculations supply the centres and fixed preconditioners, while directed interval arithmetic verifies the inclusion, contraction, first-event and multiplier inequalities.

## C The six orbits at three scales

Figure 2 shows the six cycles at three spatial scales, reconstructed from the affine flight data. Each panel uses equal scales on the coordinate axes, and all cycles retain their original physical coordinates.

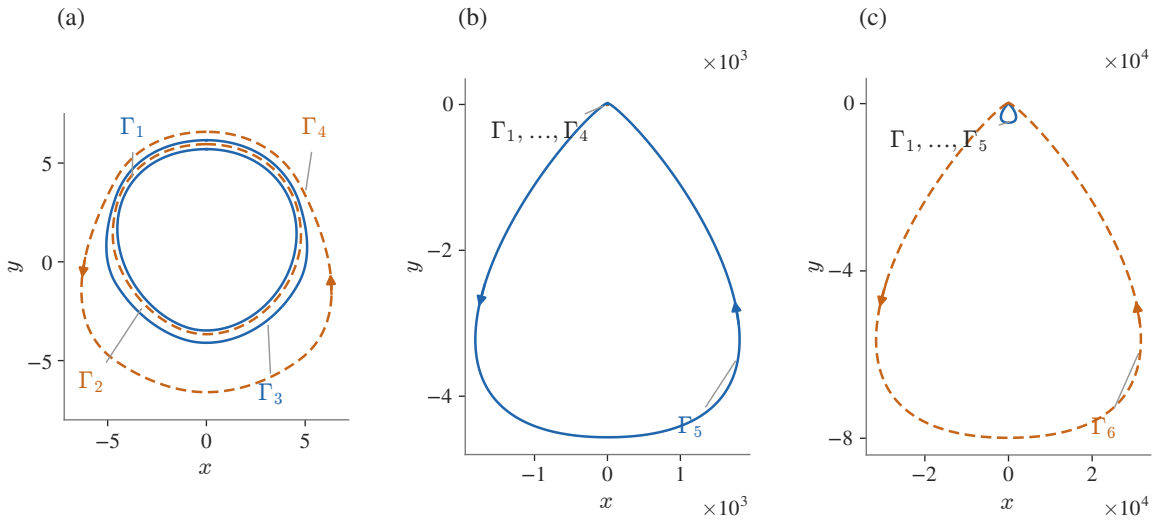



Figure 2: Numerical reconstruction of the six certified cycles from the affine flight data. The panels display successively larger physical scales; attracting and repelling cycles use solid blue and dashed orange curves.

## Declaration on AI-assisted tools

The author used OpenAI Codex for research and writing assistance. The author reviewed all content, made the final editorial decisions, and takes full responsibility for the content of the article.

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